



DOUBLY LINKED LIST

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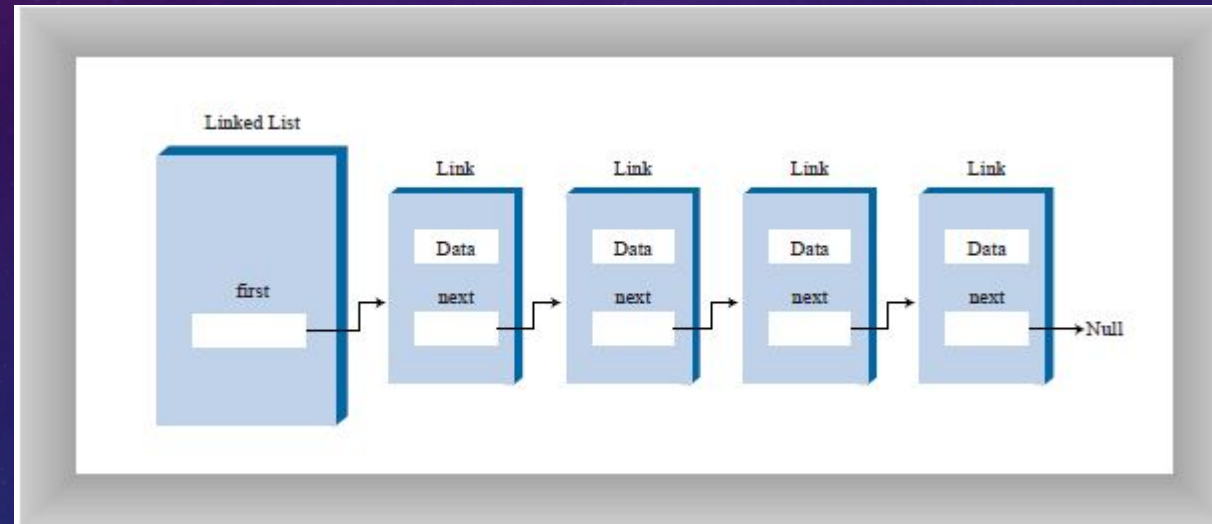
1507014- SALIM SHADMAN ANKUR

1507015- SAIFUR RAHMAN

THE CONCEPT OF THE LINKED LIST

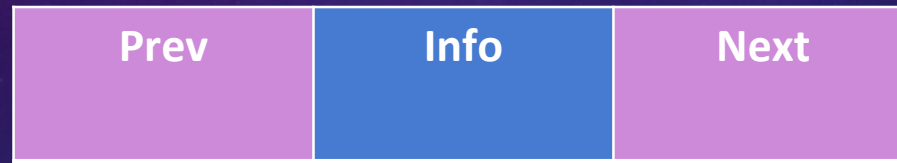
When dealing with many problems we need a dynamic list,

LINKED LIST

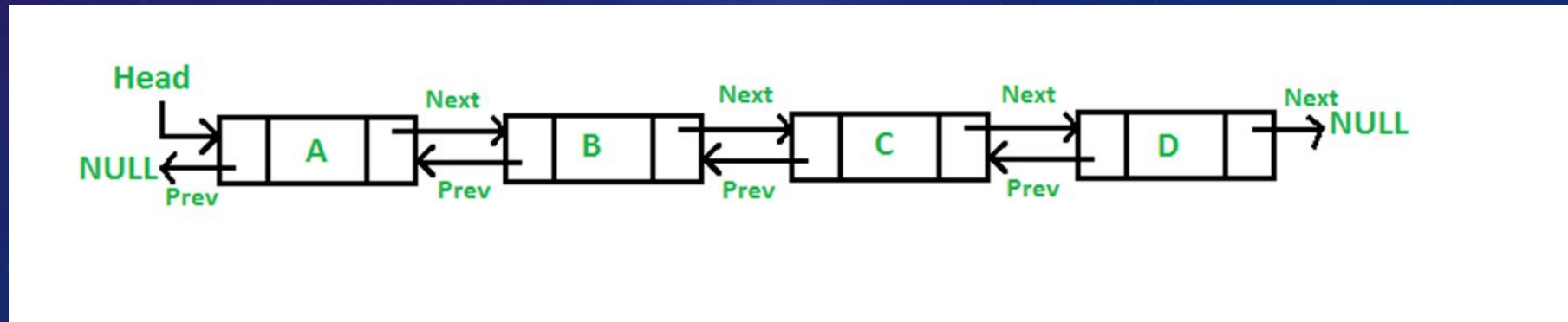


DOUBLY LINKED LIST

- Prev - address of the previous node
- Next - address of the next node
- Info - information of the node
- head - address of the first node



node



CREATION OF DOUBLY LINKED LIST

Steps-

1: Head := NULL (as global variable)

2: New Node (Ptr)

3: If Head = NULL

- Head := ptr

- temp := ptr

4. If not

- temp → next := ptr

- Ptr → prev := temp

- temp := ptr

CREATION OF DOUBLY LINKED LIST

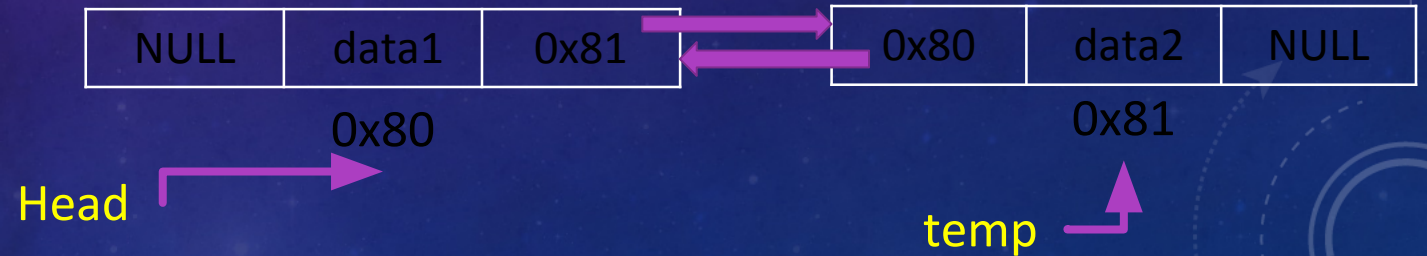
No element inserted

→ Empty list

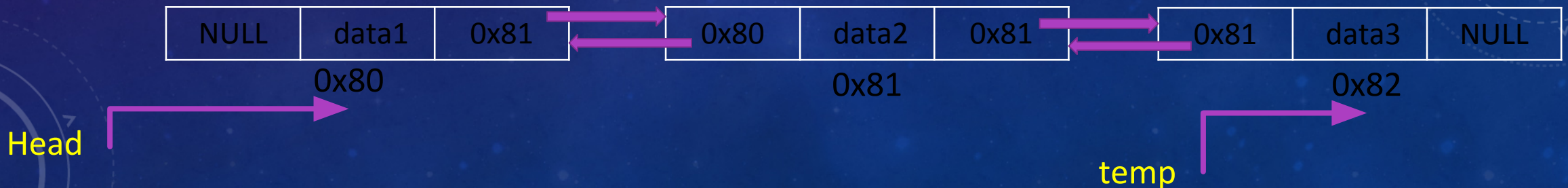
First element inserted:



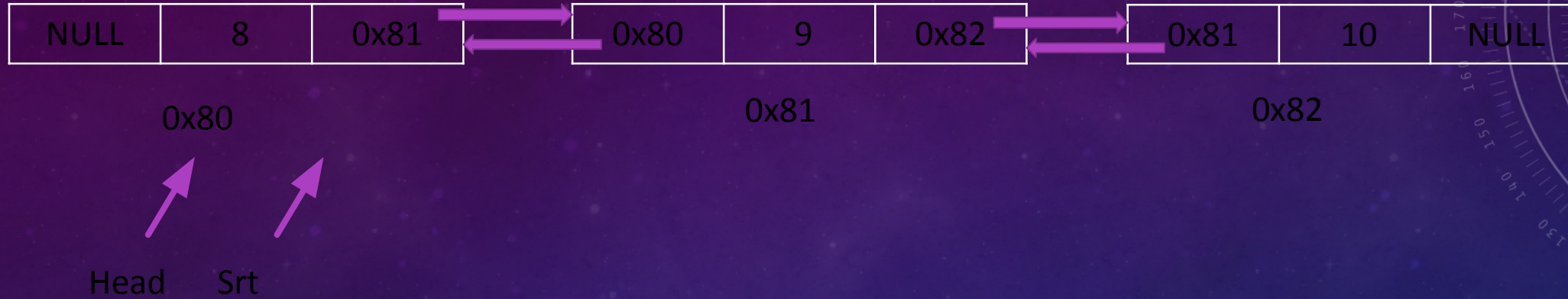
Second element inserted:



Third element inserted:



TRAVERSING



Steps:

Set srt := head

while (srt->next != NULL)

print srt->info // 8

srt := srt->next

TRAVERSING



Steps:

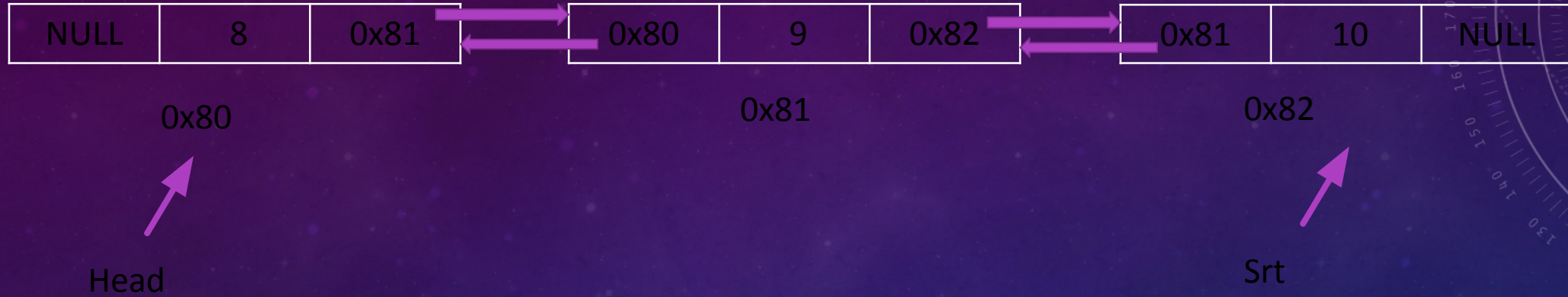
Set srt := head

while (srt->next != NULL)

print srt->info // 9

srt := srt->next

TRAVERSING



Steps:

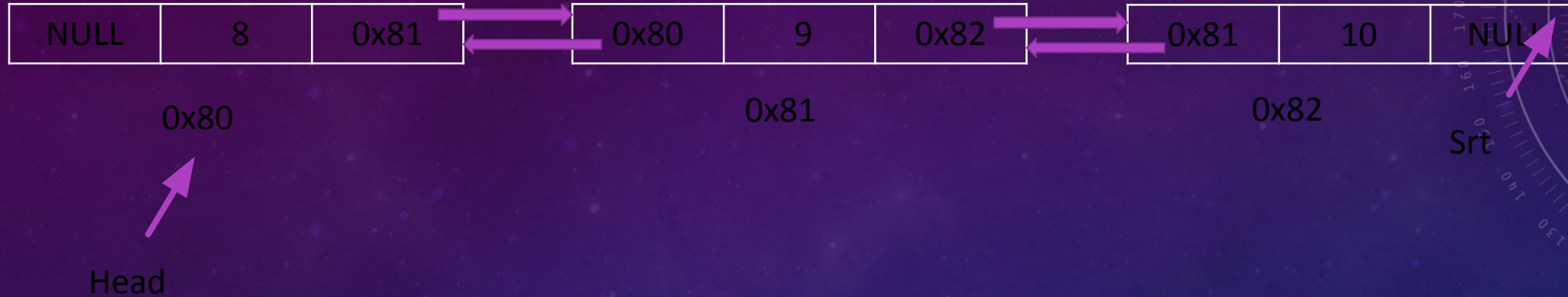
Set $srt := head$

while ($srt \rightarrow next \neq NULL$)

print $srt \rightarrow info$ // 10

$srt := srt \rightarrow next$

TRAVERSING



Steps:

Set srt := head

while (srt->next != NULL) //As srt->next := NULL ,traversing is finished

print srt->info

srt := srt->next

OPERATIONS ON DOUBLY LINKED LIST

Insertion

- Insert an element at first
- Insert an element at last
- Insert before a given element
- Insert after a given element

Deletion

- Delete first element
- Delete last element
- Delete a given element
- Delete before a given element
- Delete after a given element

INSERT AN ELEMENT AT FIRST



Steps-

1. New node (Ptr)

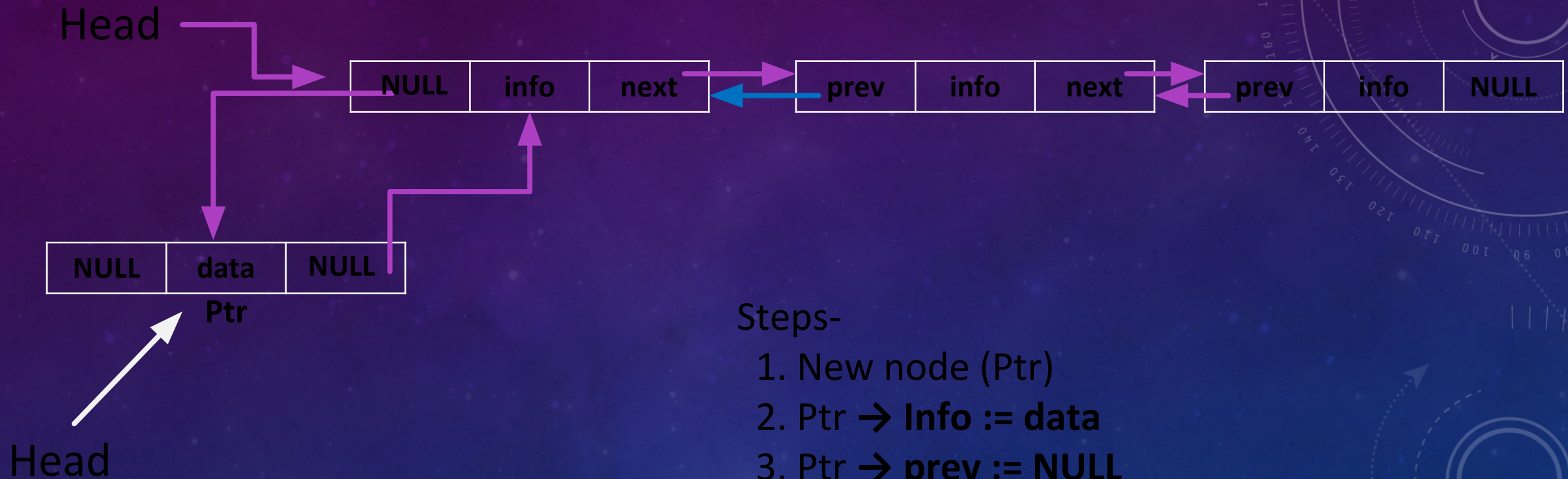
INSERT AN ELEMENT AT FIRST



Steps-

1. New node (Ptr)
2. $\text{Ptr} \rightarrow \text{info} := \text{data}$
3. $\text{Ptr} \rightarrow \text{prev} := \text{NULL}$
4. $\text{Ptr} \rightarrow \text{next} := \text{NULL}$

INSERT AN ELEMENT AT FIRST



Steps-

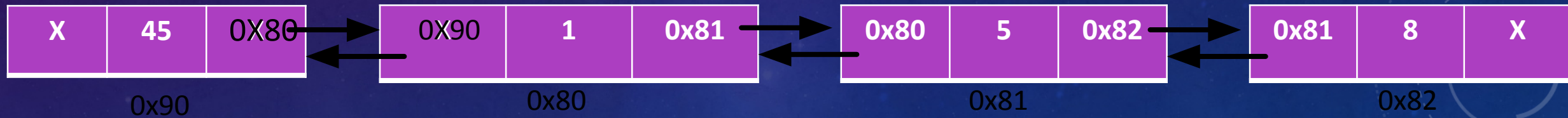
1. New node (Ptr)
2. $\text{Ptr} \rightarrow \text{Info} := \text{data}$
3. $\text{Ptr} \rightarrow \text{prev} := \text{NULL}$
4. $\text{Ptr} \rightarrow \text{next} := \text{NULL}$
5. $\text{Head} \rightarrow \text{prev} := \text{ptr}$
6. $\text{Ptr} \rightarrow \text{next} := \text{head}$
7. $\text{Head} := \text{Ptr}$

EXAMPLE OF FIRST INSERTION:

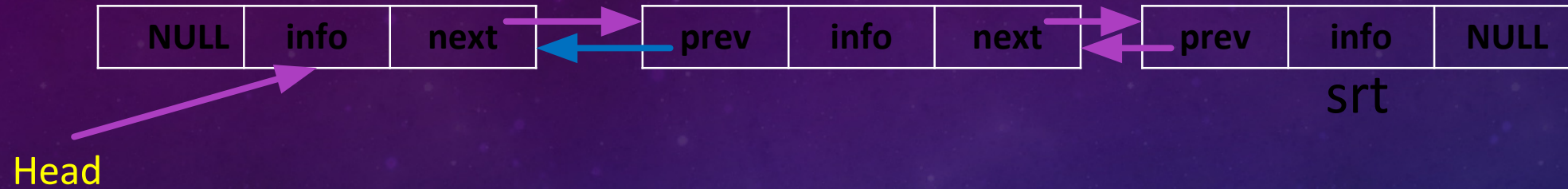
This is a doubly linked list



Inserting 45 at the first of the list

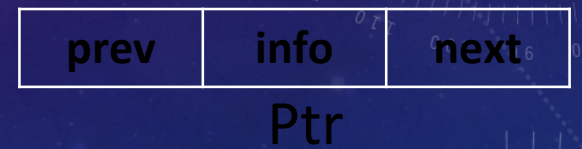


INSERT AN ELEMENT AT LAST

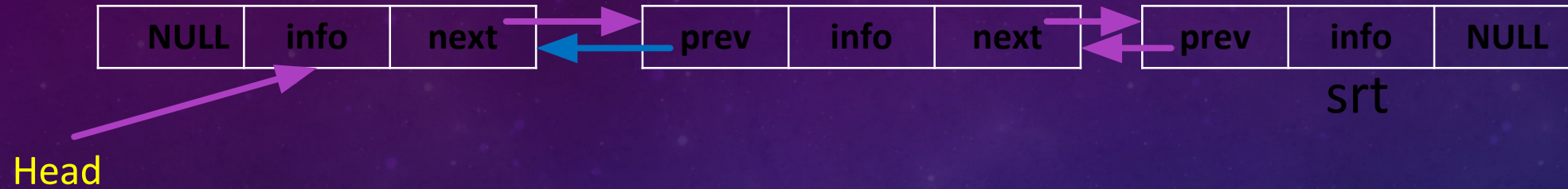


Steps-

1. New node (Ptr)



INSERT AN ELEMENT AT LAST



Steps-

1. New node (Ptr)
2. $\text{Ptr} \rightarrow \text{Info} := \text{data}$



INSERT AN ELEMENT AT LAST



Steps-

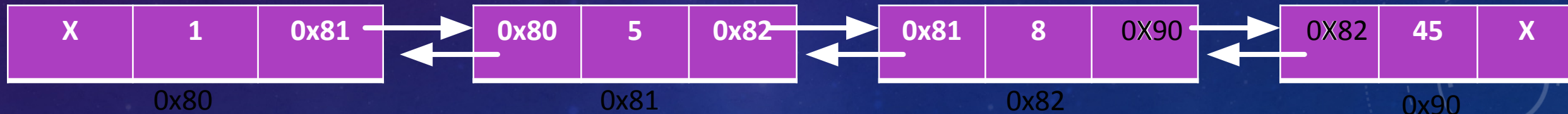
1. New node (Ptr)
2. $\text{Ptr} \rightarrow \text{Info} := \text{data}$
3. $\text{Ptr} \rightarrow \text{prev} := \text{NULL}$
4. $\text{Ptr} \rightarrow \text{next} := \text{NULL}$
5. $\text{srt} \rightarrow \text{next} := \text{ptr}$
6. $\text{Ptr} \rightarrow \text{prev} := \text{srt}$

EXAMPLE OF LAST INSERTION:

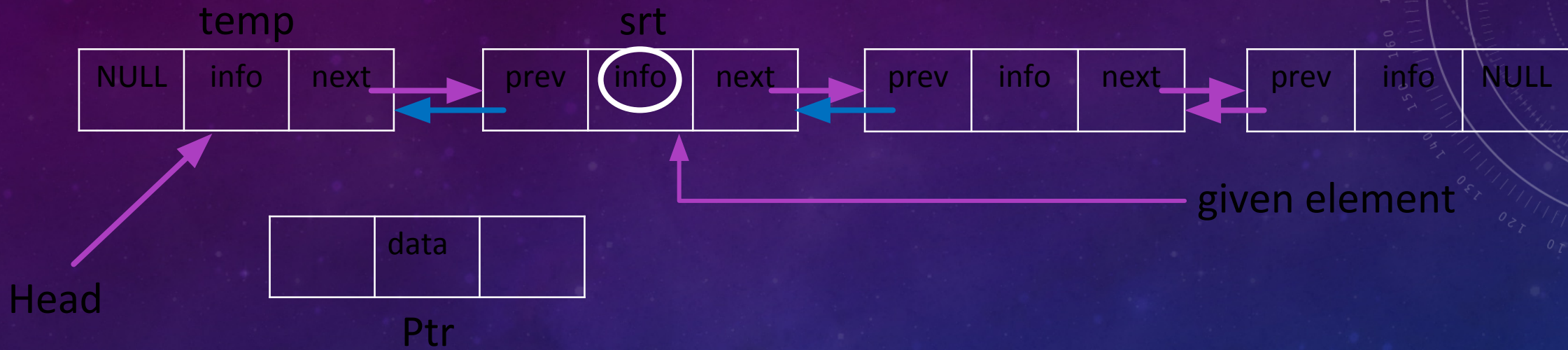
This is a doubly linked list Insert 45 at the last of the list



After insertion 45 at last....



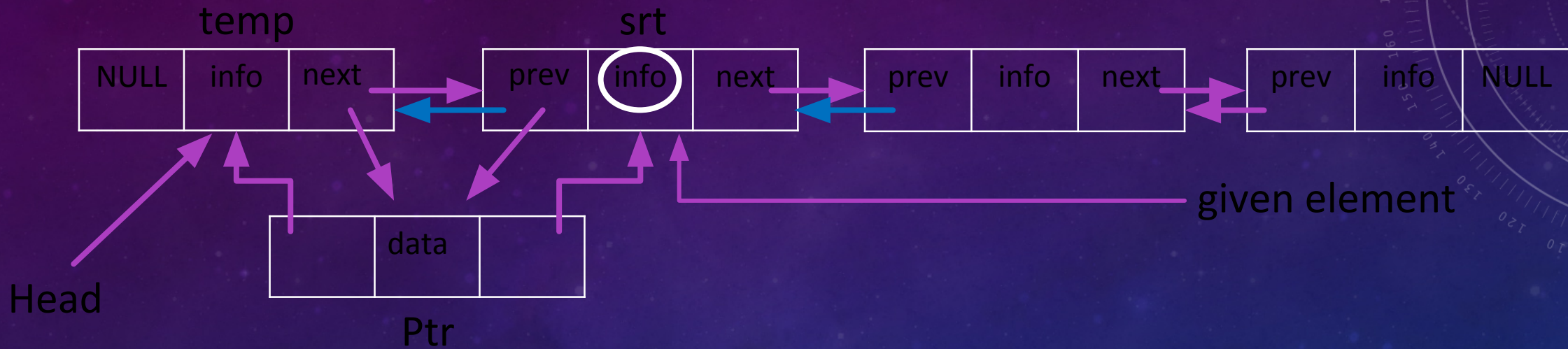
INSERT BEFORE A GIVEN ELEMENT



Steps-

1. New Node(Ptr)
2. ptr → info = data

INSERT BEFORE A GIVEN ELEMENT



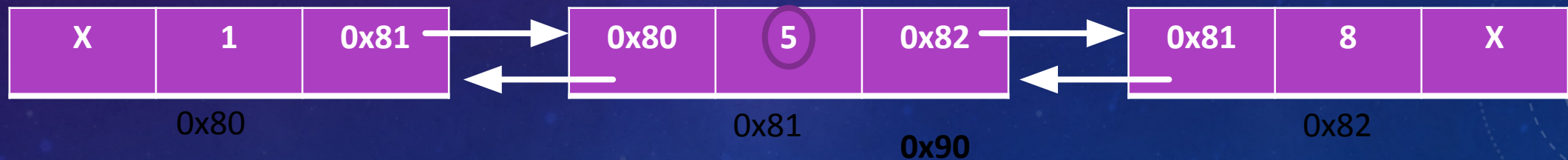
Steps-

1. New Node(`Ptr`)
2. `ptr` $\rightarrow$ `info` = `data`
3. `srt` $\rightarrow$ `prev` := `Ptr`
4. `temp` $\rightarrow$ `next` := `Ptr`
5. `ptr` $\rightarrow$ `prev` := `temp`
6. `ptr` $\rightarrow$ `next` := `srt`

EXAMPLE INSERT BEFORE A GIVEN ELEMENT:



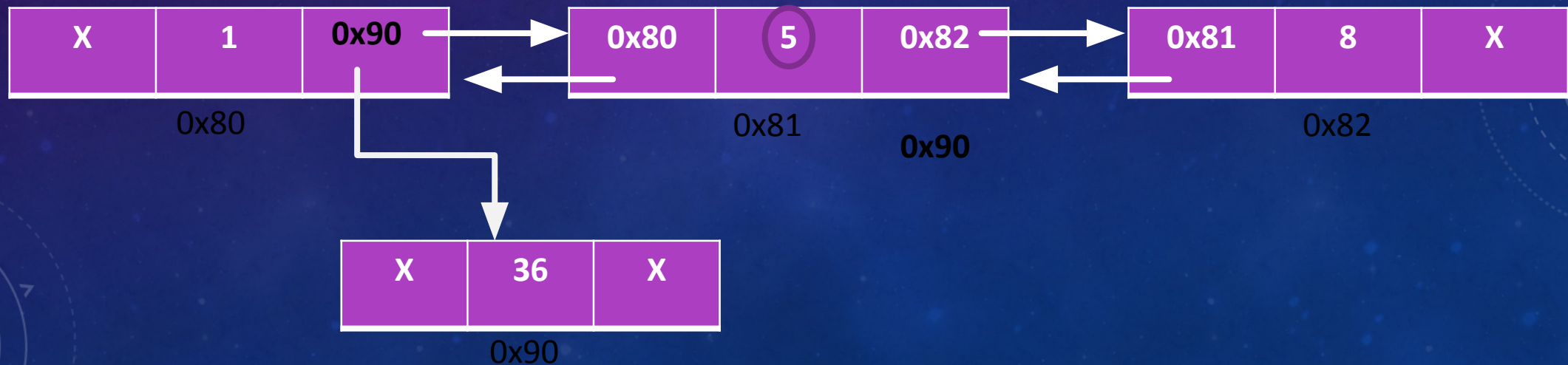
Inserting 36 before 5...



EXAMPLE INSERT BEFORE A GIVEN ELEMENT:



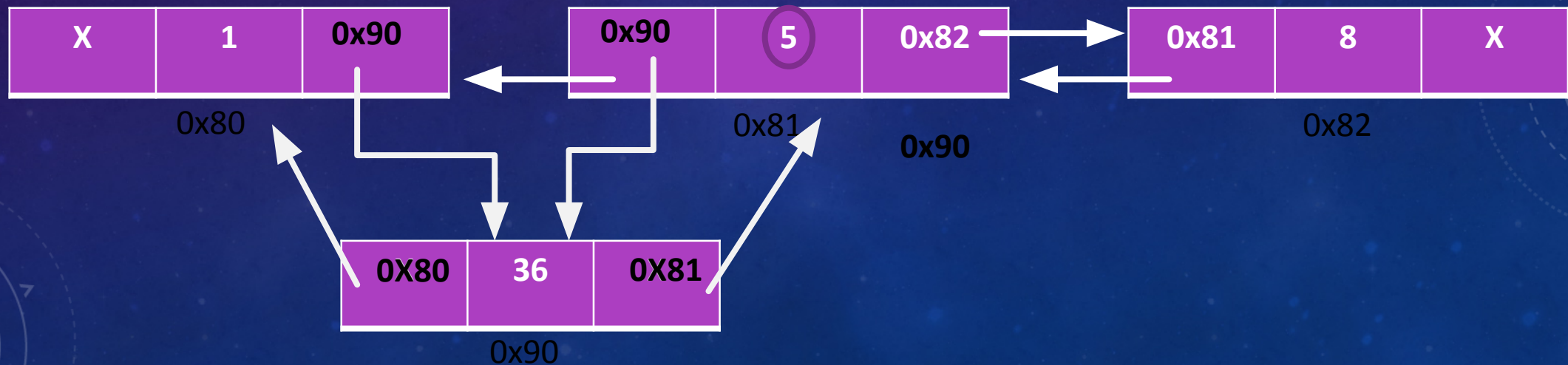
Inserting 36 before 5...



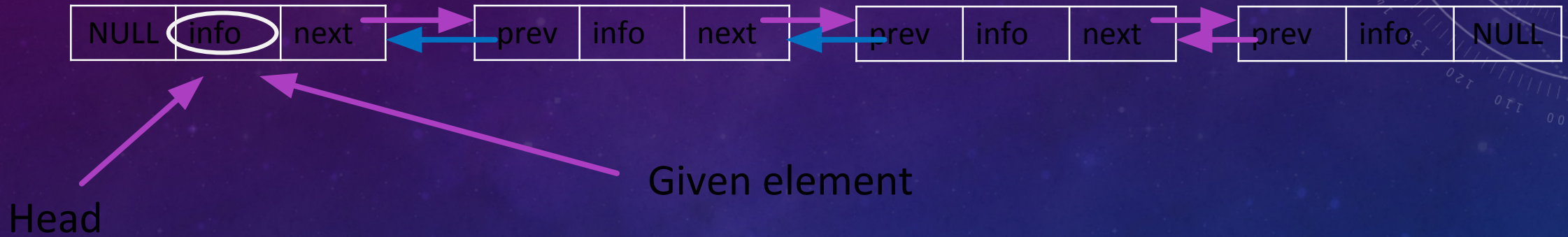
EXAMPLE INSERT BEFORE A GIVEN ELEMENT:



Inserting 36 before 5...

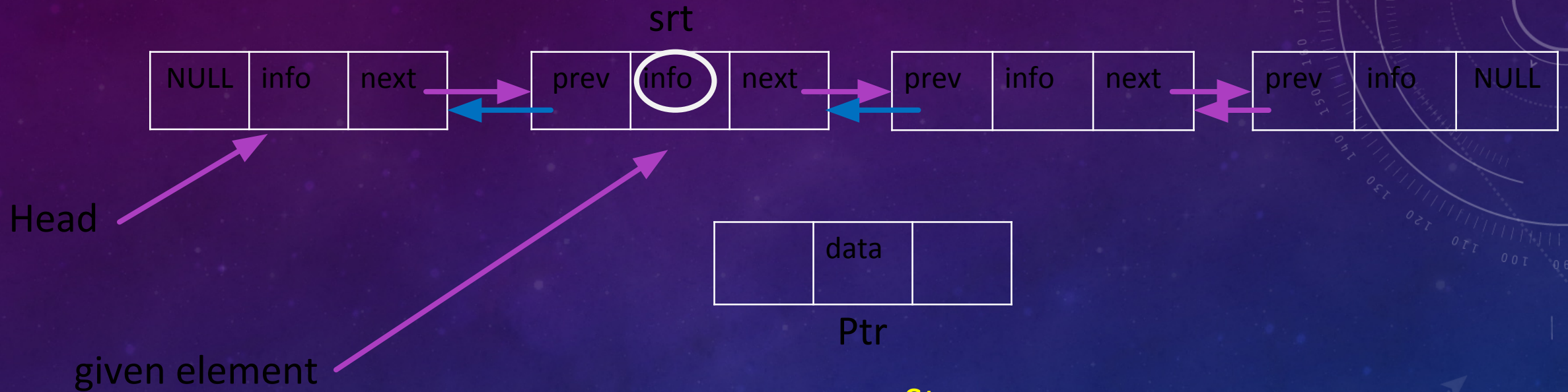


SPECIAL CASE: (IF GIVEN ELEMENT IS THE FIRST NODE OF THE LIST)



- ❖ When Head $\rightarrow$ info = given element
- In this case it is similar to first insertion

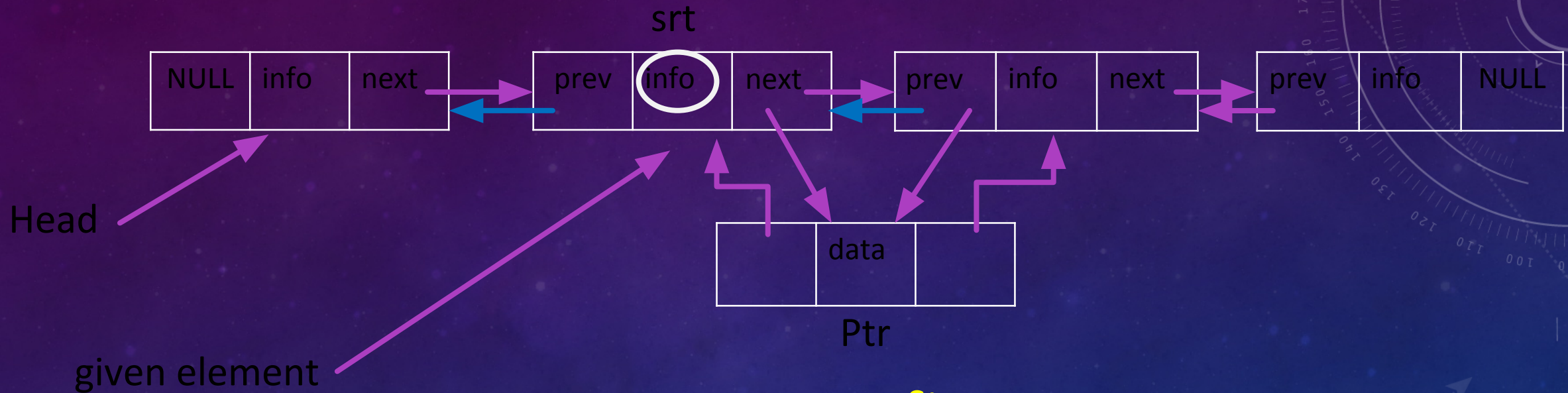
INSERT AFTER A GIVEN ELEMENT



Steps-

1. New Node (Ptr)
2. $\text{Ptr} \rightarrow \text{info} := \text{data}$

INSERT AFTER A GIVEN ELEMENT



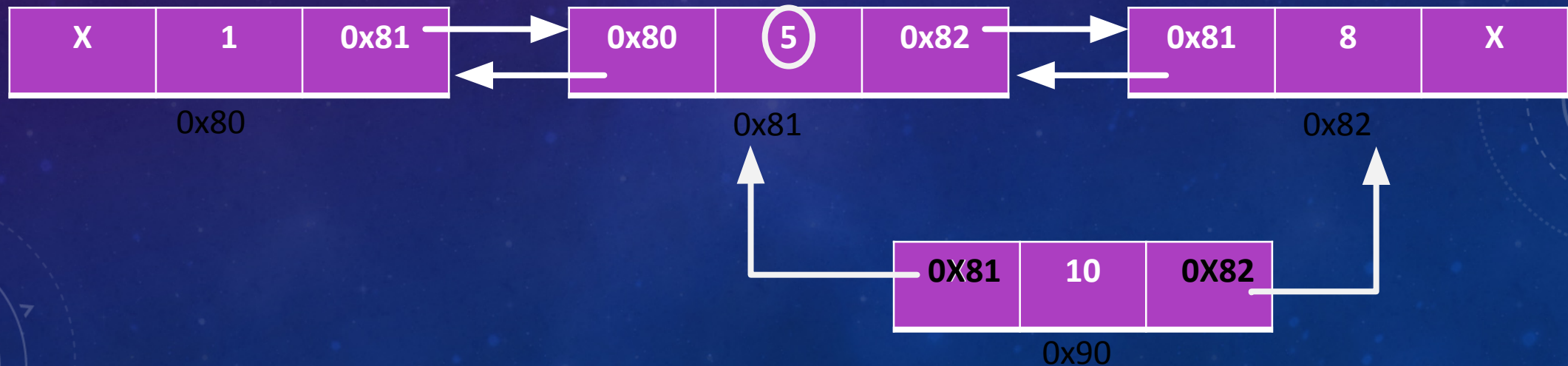
Steps-

1. New Node (Ptr)
2. $\text{Ptr} \rightarrow \text{info} := \text{data}$
3. $\text{Ptr} \rightarrow \text{prev} := \text{srt}$
4. $\text{Ptr} \rightarrow \text{next} := \text{srt} \rightarrow \text{next}$
5. $\text{srt} \rightarrow \text{next} \rightarrow \text{prev} := \text{Ptr}$
6. $\text{srt} \rightarrow \text{next} := \text{Ptr}$

EXAMPLE INSERT AFTER A GIVEN ELEMENT:



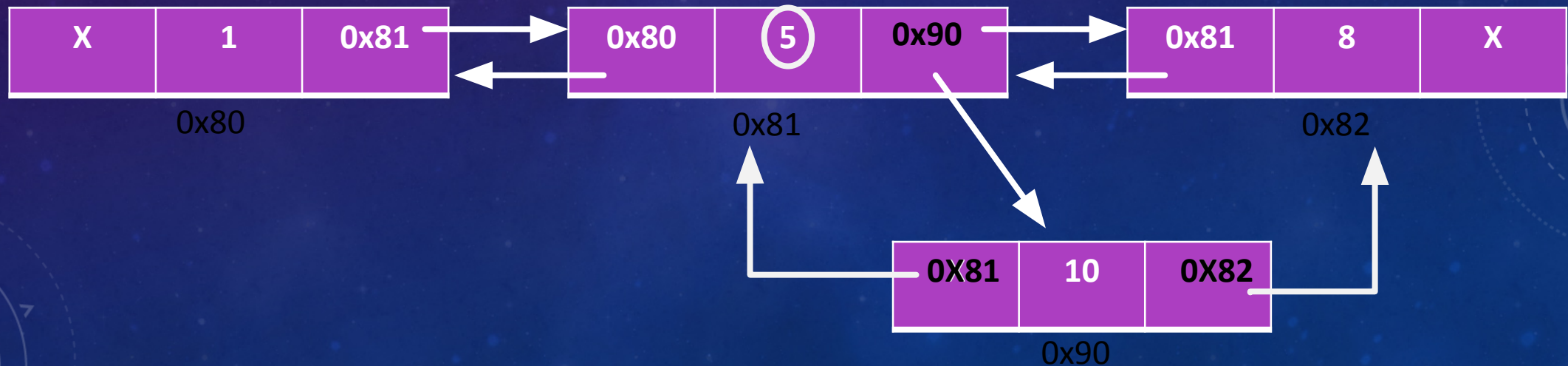
Inserting 10 after 5...



EXAMPLE INSERT AFTER A GIVEN ELEMENT:



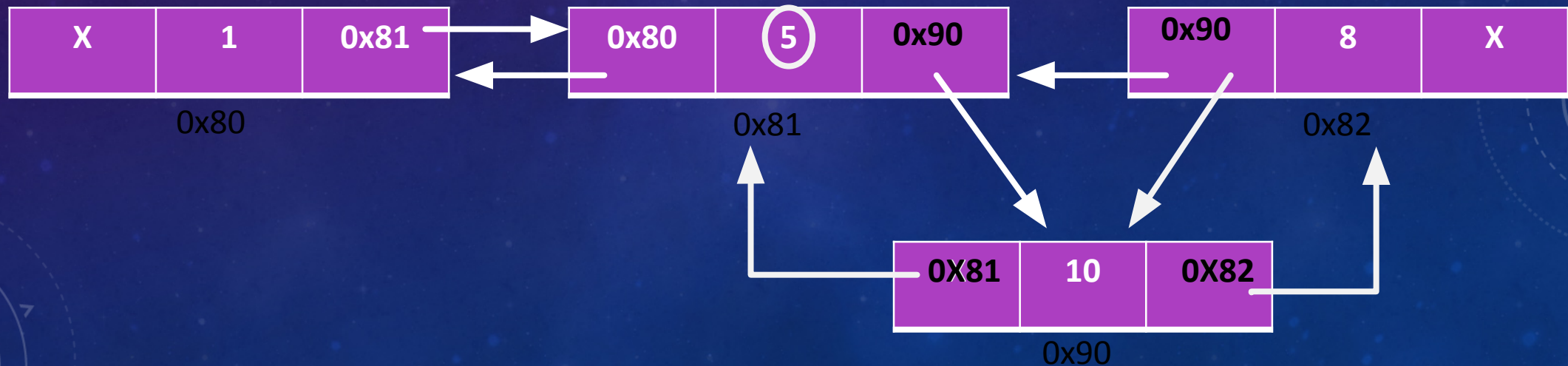
Inserting 10 after 5...



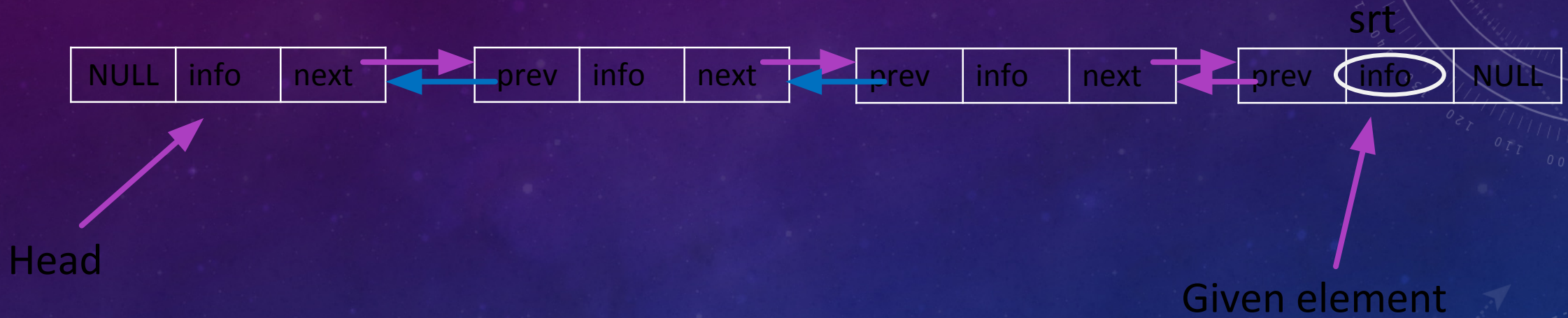
EXAMPLE INSERT AFTER A GIVEN ELEMENT:



Inserting 10 after 5...



SPECIAL CASE: (IF GIVEN ELEMENT IS THE LAST NODE OF THE LIST)

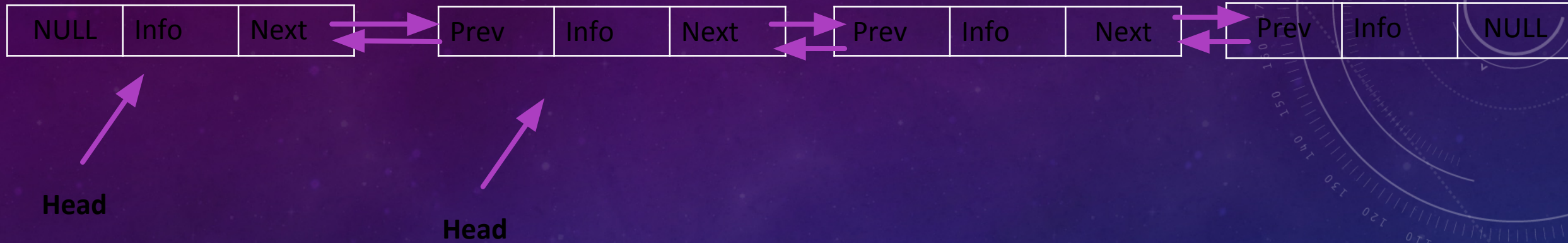


- ❖ When `srt` $\rightarrow$ `next` = NULL
- In this case it is similar to last insertion



**Now It's Time for
Deletion**

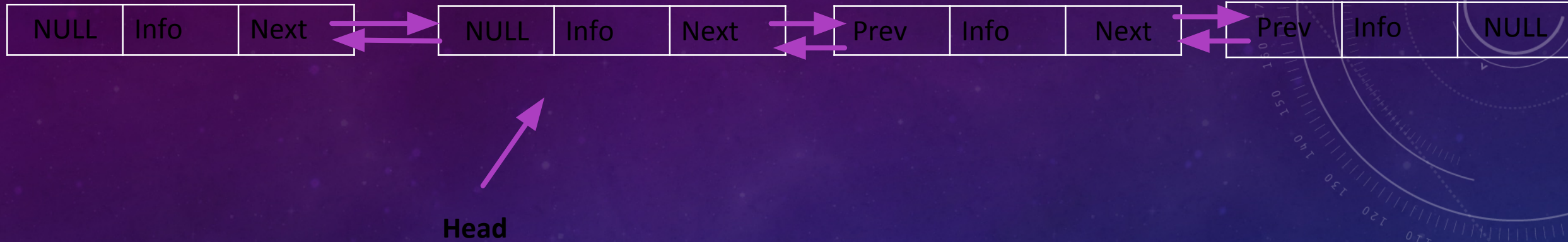
DELETE AN ELEMENT AT FIRST



Steps-

1. Set head := head->next
2. Set head->prev := NULL

DELETE AN ELEMENT AT FIRST

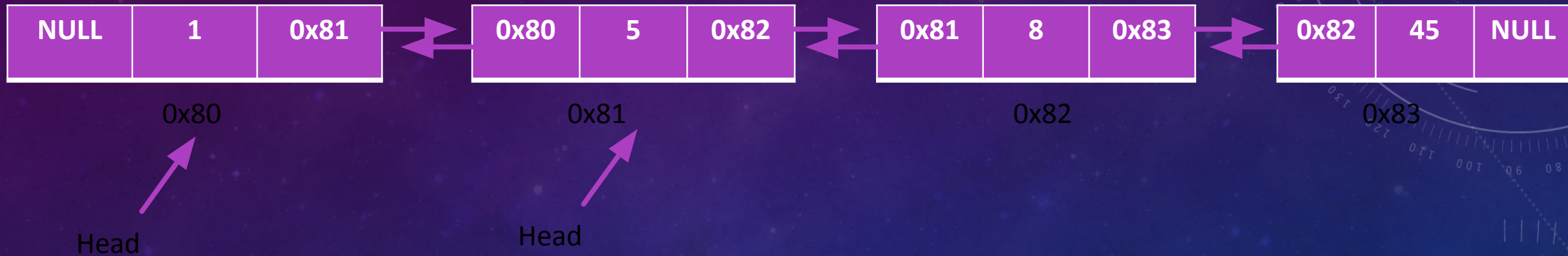


Steps-

1. Set $\text{head} := \text{head} \rightarrow \text{next}$
2. Set $\text{head} \rightarrow \text{prev} := \text{NULL}$

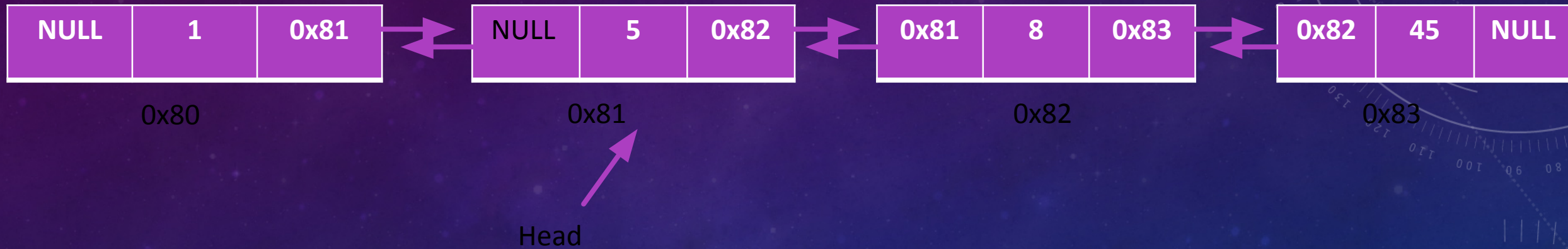
EXAMPLE OF FIRST DELETION:

This is a doubly linked list Delete the first node..

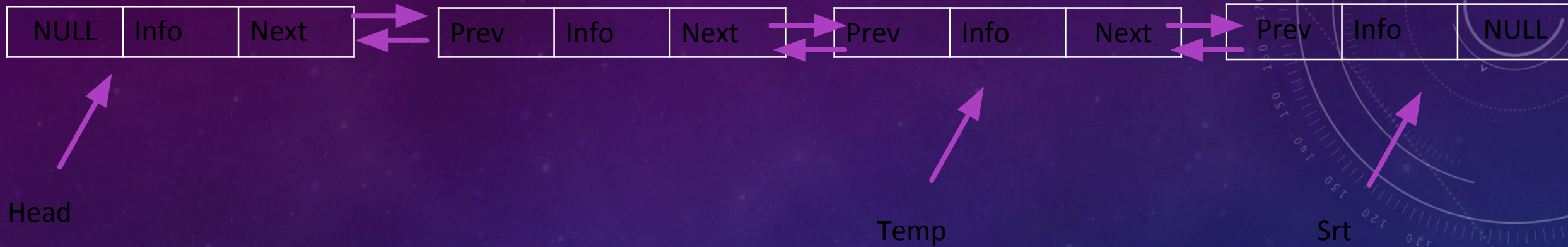


EXAMPLE OF FIRST DELETION:

After deleting first node



DELETE AN ELEMENT AT LAST

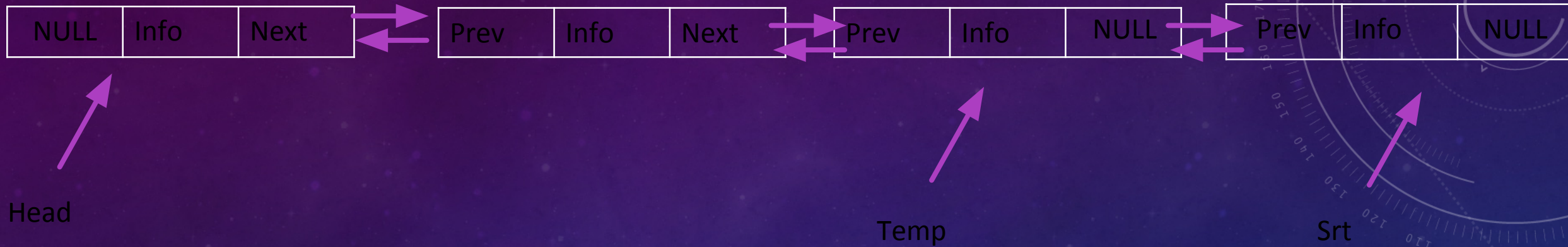


Steps-

When $\text{srt} \rightarrow \text{next} := \text{NULL}$

1. Set $\text{temp} \rightarrow \text{next} := \text{NULL}$

DELETE AN ELEMENT AT LAST



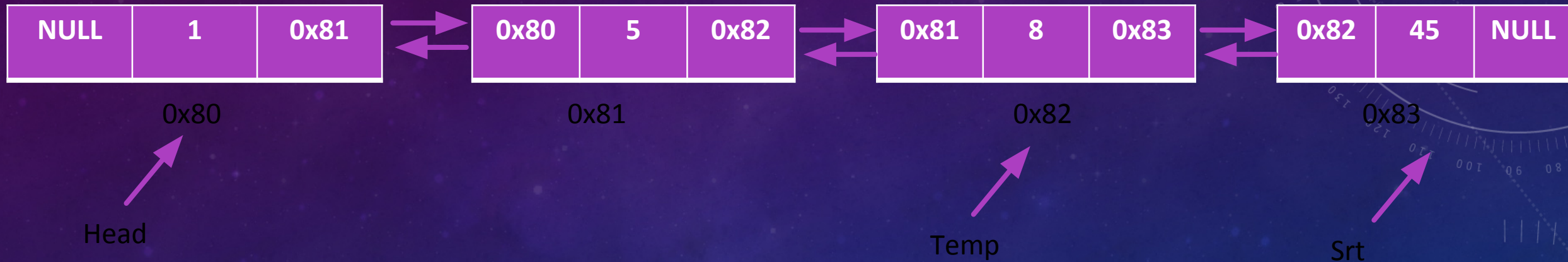
Steps-

When $\text{srt} \rightarrow \text{next} := \text{NULL}$

1. Set $\text{temp} \rightarrow \text{next} := \text{NULL}$

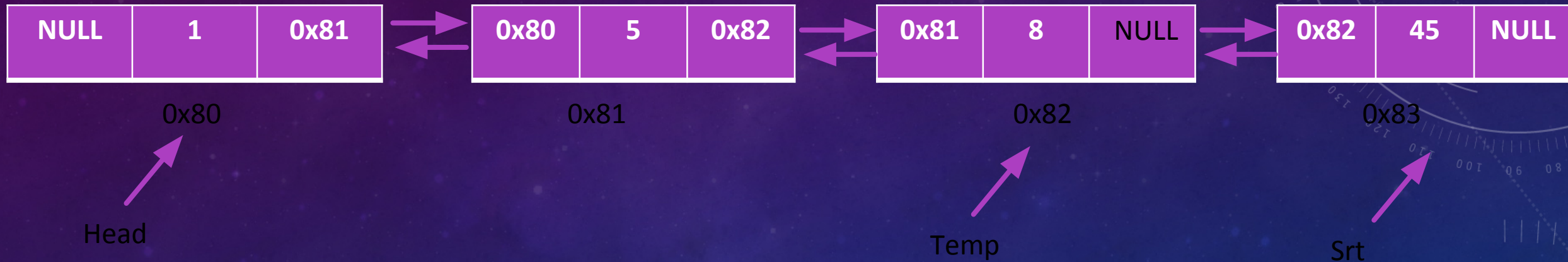
EXAMPLE OF LAST DELETION:

This is a doubly linked list Delete the last node..

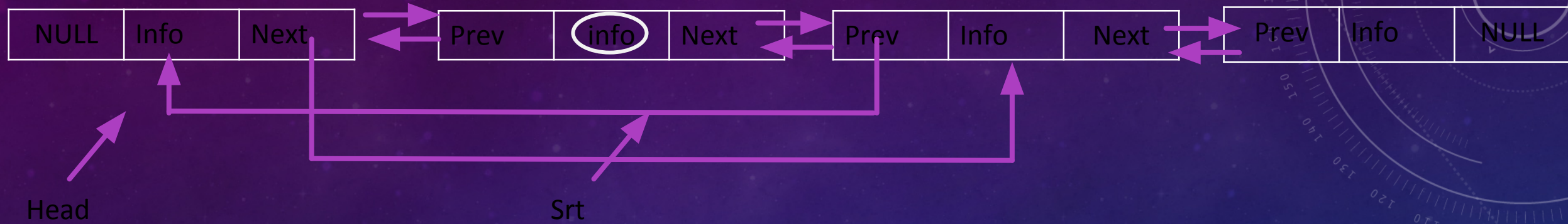


EXAMPLE OF LAST DELETION:

This is a doubly linked list Delete the last node..



DELETE A NODE OF GIVEN ELEMENT



Steps-

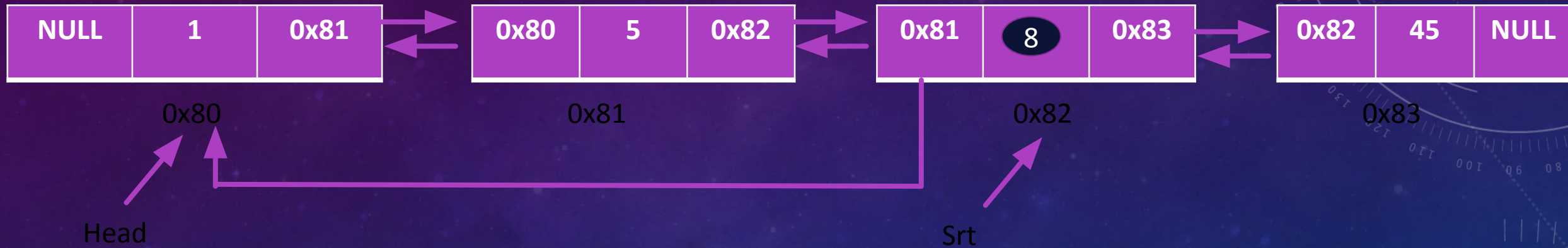
when $\text{srt} \rightarrow \text{info} := \text{item}$

$\text{srt} \rightarrow \text{next} \rightarrow \text{prev} := \text{srt} \rightarrow \text{prev}$

$\text{srt} \rightarrow \text{prev} \rightarrow \text{next} := \text{srt} \rightarrow \text{next}$

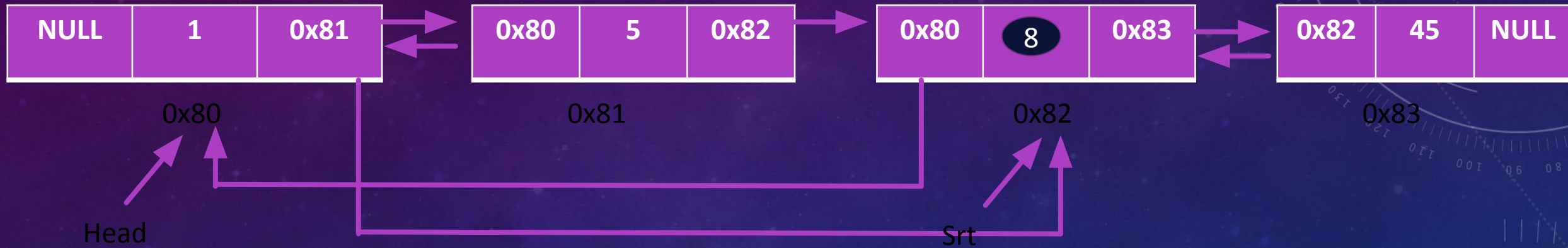
EXAMPLE OF DELETION OF A GIVEN ELEMENT:

This is a doubly linked list Delete the particular node.. Suppose the item is 8



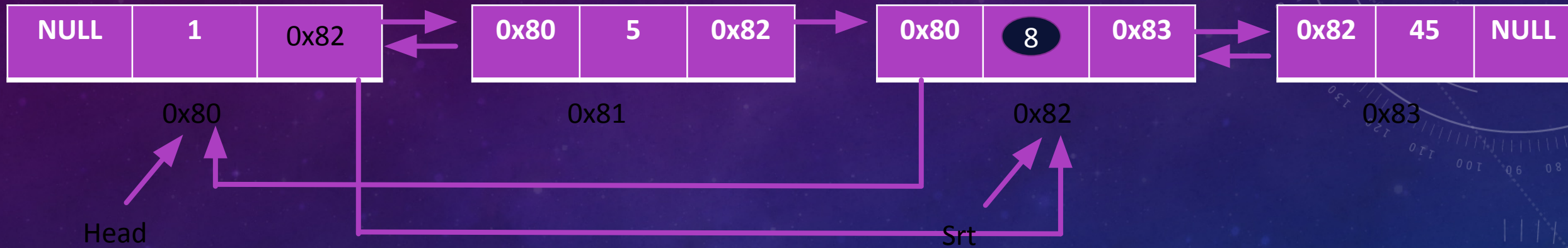
EXAMPLE OF DELETION OF A GIVEN ELEMENT:

This is a doubly linked list Delete the particular node.. Suppose the item is 8



EXAMPLE OF DELETION OF A GIVEN ELEMENT:

After deleting the particular node

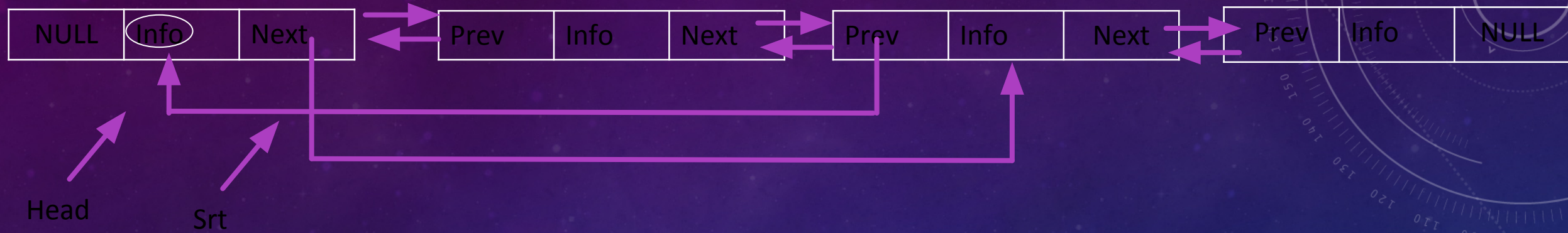


SPECIAL CASES: FOR DELETION OF GIVEN ELEMENT

I) If the given element is in the first node, it will be as same as first element delete.

II) If the given element is in the last node, it will be as same as last element delete.

DELETE A NODE AFTER GIVEN ELEMENT



Steps-

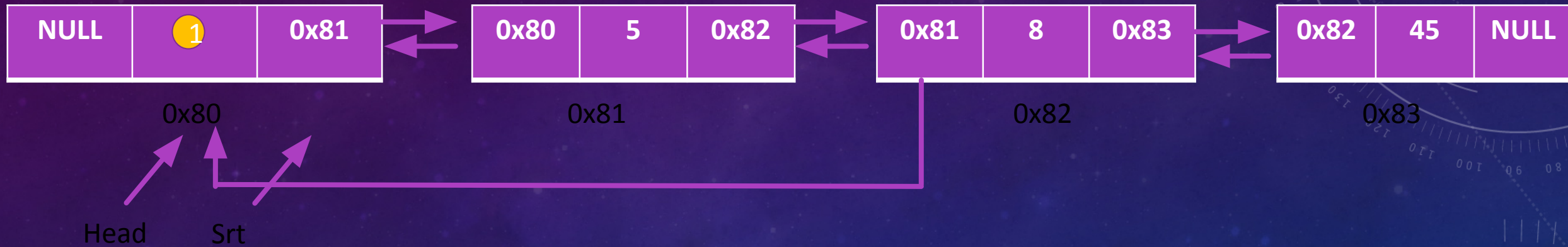
when $\text{srt} \rightarrow \text{info} := \text{item}$

Set $\text{srt} \rightarrow \text{next} \rightarrow \text{next} \rightarrow \text{prev} := \text{srt}$

$\text{srt} \rightarrow \text{next} = \text{srt} \rightarrow \text{next} \rightarrow \text{next}$

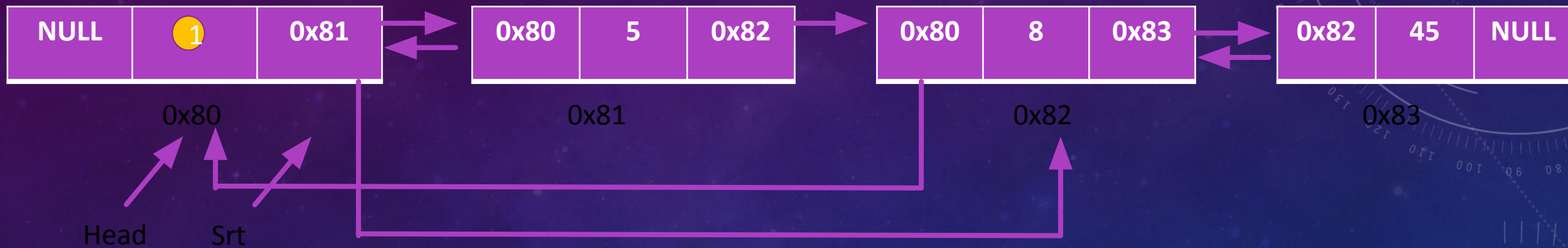
EXAMPLE OF DELETION AFTER A GIVEN ELEMENT:

This is a doubly linked list Delete the particular node.. Suppose the item is 8



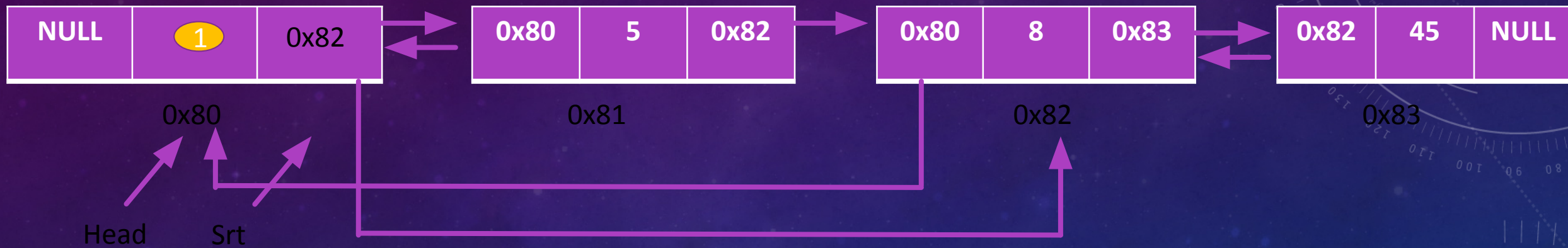
EXAMPLE OF DELETION AFTER A GIVEN ELEMENT:

This is a doubly linked list Delete the particular node.. Suppose the item is 8



EXAMPLE OF DELETION AFTER A GIVEN ELEMENT:

Deleting the particular node after the given element

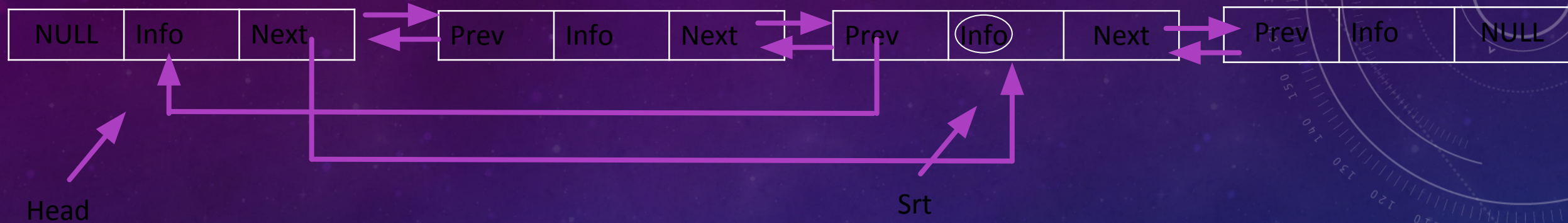


SPECIAL CASES: FOR DELETION AFTER GIVEN ELEMENT

I) If the given element is before the last node , it will be as same as last element delete.

II) If the given element is in the last node, it will not be possible. As the next of last node is NULL , it will not delete anything

DELETE A NODE BEFORE GIVEN ELEMENT



Steps-

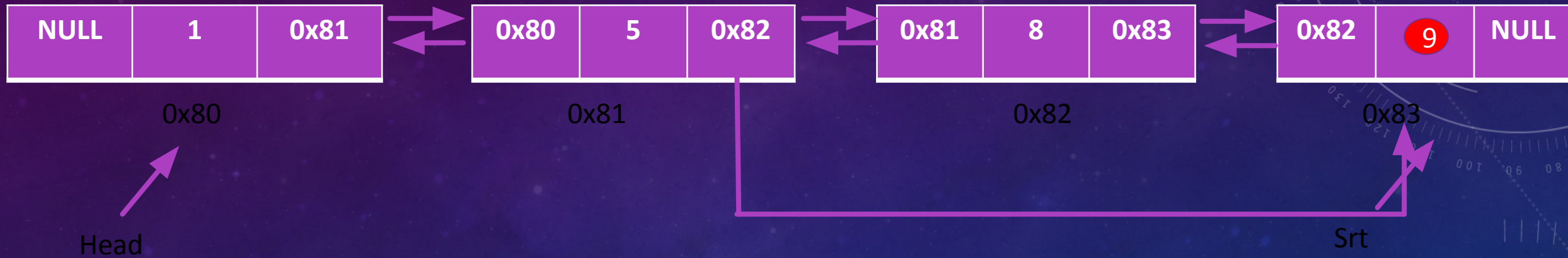
when $srt \rightarrow info := item$

Set $srt \rightarrow prev \rightarrow prev \rightarrow next := srt$

$srt \rightarrow prev := srt \rightarrow prev \rightarrow prev$

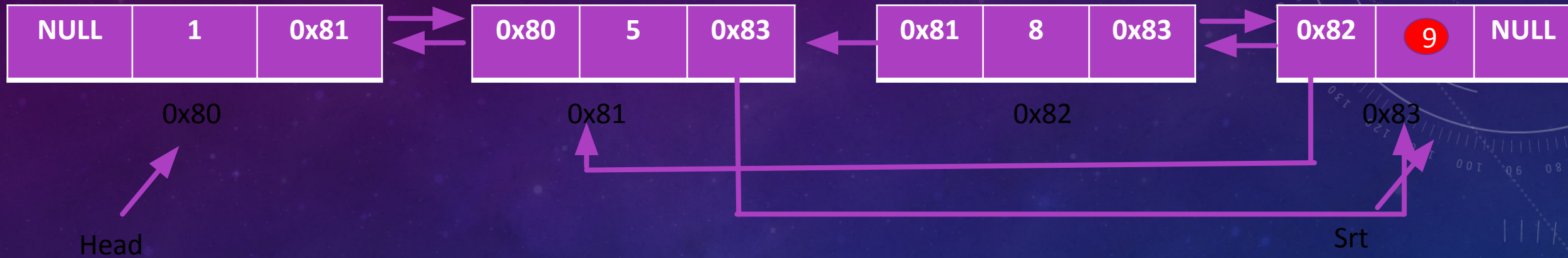
EXAMPLE OF DELETION BEFORE A GIVEN ELEMENT:

This is a doubly linked list Delete the particular node before a given item.. Suppose the item is 9



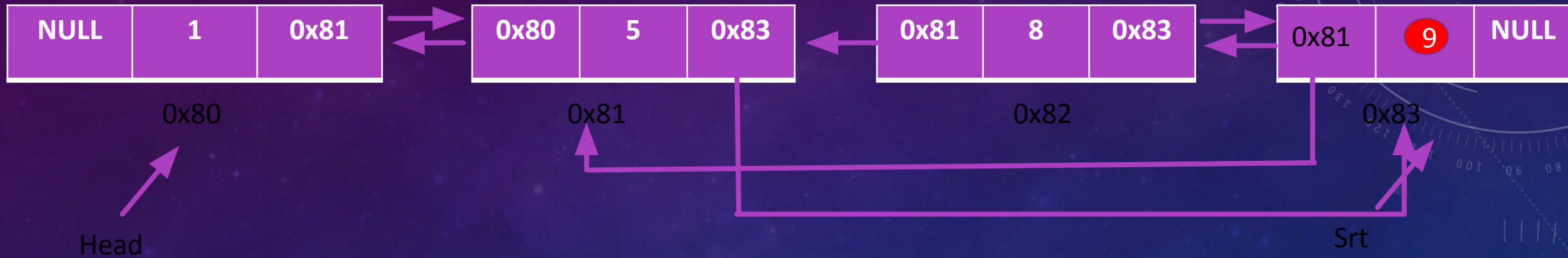
EXAMPLE OF DELETION BEFORE A GIVEN ELEMENT:

This is a doubly linked list Delete the particular node before a given item.. Suppose the item is 9



EXAMPLE OF DELETION BEFORE A GIVEN ELEMENT:

After Deleting The Node Before A Given Element



SPECIAL CASES: FOR DELETION AFTER GIVEN ELEMENT

I) If the given element is after the first node , it will be as same as first element delete.

II) If the given element is in the first node, it will not be possible. As the next of first node is NULL , it will not delete anything

ADVANTAGES OF DOUBLY LINKED LIST

- 1) A DLL can be traversed in both forward and backward direction.
- 2) The delete operation in DLL is more efficient if pointer to the node to be deleted is given.
In singly linked list, to delete a node, pointer to the previous node is needed. To get this previous node, sometimes the list is traversed. In DLL, we can get the previous node using previous pointer.

DISADVANTAGES OF DOUBLY LINKED LIST

- 1) Every node of DLL Require extra space for an previous pointer. It is possible to implement DLL with single pointer though .
- 2) All operations require an extra pointer previous to be maintained. For example, in insertion, we need to modify previous pointers together with next pointers. For example in following functions for insertions at different positions, we need 1 or 2 extra steps to set previous pointer